

Health Spending and Life Expectancy in Eighteen OECD Countries

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August 24, 2026

Introduction

I want to produce a version of a figure I first saw in Kenworthy (2014, 51). Versions of it have appeared elsewhere, too. To make it we'll need to get data from the OECD and then write some code to draw the graph.

```
library(tidyverse)
library(tinytable)
my_colors <- c("grey50", "firebrick")
```

The Data

We're working in this little project, so our local data files and our output is defined with respect to where the project is on our computer. In R, the **here** package helps us stay disciplined about this.

```
library(here)
```

We set things up by getting the data from a file in the project.¹ It's a comma-separated values or CSV file. To do our work we'll put it in a thing named `df`. It looks like this:

```
## The data are generated by R/make_oecd_df.R
df <- read_csv(
  here("data", "oecd_health_lifexp.csv"),
  col_types = cols(
```

¹Don't worry at this point if you don't know any R.

```

    country = col_character(),
    iso3 = col_character(),
    year = col_integer(),
    life_exp = col_double(),
    health_ppp = col_double()
  )
)

```

```
df
```

```

# A tibble: 2,238 x 5
  country iso3 year life_exp health_ppp
  <chr>   <chr> <int>   <dbl>   <dbl>
1 Australia AUS   1962    71      NA
2 Australia AUS   1967   70.8    NA
3 Australia AUS   1971    NA     999.
4 Australia AUS   1972    NA    1027.
5 Australia AUS   1973    NA    1080.
6 Australia AUS   1974    NA    1199.
7 Australia AUS   1975    NA    1351.
8 Australia AUS   1976   72.8    1387.
9 Australia AUS   1977    NA    1444.
10 Australia AUS   1978    NA    1451.
# i 2,228 more rows

```

```

my_countries <- c("AUT", "AUS", "BEL", "CAN", "DEU", "DNK", "ESP",
                  "FIN", "FRA", "GBR", "GRC", "IRL", "ITA", "JPN",
                  "NLD", "NOR", "NZL", "SWE", "USA")
cnames <- df$country[df$iso3 %in% my_countries] |>
  unique() |>
  paste0(collapse = ", ")

cnames <- str_replace(cnames, "United States", "and the United States")

```

There's more data here than we are interested in. We'll look at these countries only: Australia, Austria, Belgium, Canada, Germany, Denmark, Spain, Finland, France, United Kingdom, Greece, Ireland, Italy, Japan, Netherlands, Norway, New Zealand, Sweden, and the United States. We're also just interested in 1970 and after. And in particular we want to draw a figure that contrasts the US and all the other countries. For that we'll make an indicator or flag or dummy variable that picks out the US from all the other countries. Finally, we'll smooth the trends a little by calculating a five-year moving average for each country.

```
df_plot <- df |>
  filter(iso3 %in% my_countries, year > 1969) |>
  drop_na() |>
  arrange(country, year) |>
  group_by(country) |>
  mutate(
    us_flag = ifelse(iso3 == "USA", "United States",
                     "Eighteen OECD Countries"),
    avg_spend = slider::slide_dbl(health_ppp, mean, .before = 2, .after = 2)
  )
```

On our computer, we end up with a data frame that looks like this:

```
df_plot

# A tibble: 909 x 7
# Groups:   country [19]
  country iso3 year life_exp health_ppp us_flag avg_spend
  <chr>   <chr> <int>   <dbl>     <dbl> <chr>     <dbl>
1 Australia AUS  1976    72.8    1387. Eighteen OECD Countries 1480.
2 Australia AUS  1981    74.8    1527. Eighteen OECD Countries 1503.
3 Australia AUS  1982    74.6    1526. Eighteen OECD Countries 1523.
4 Australia AUS  1983    75.4    1572. Eighteen OECD Countries 1576.
5 Australia AUS  1984    75.7    1606. Eighteen OECD Countries 1612.
6 Australia AUS  1985    75.5    1649. Eighteen OECD Countries 1655.
7 Australia AUS  1986    76     1706. Eighteen OECD Countries 1702.
8 Australia AUS  1987    76.2    1741. Eighteen OECD Countries 1751.
9 Australia AUS  1988    76.2    1809. Eighteen OECD Countries 1798.
10 Australia AUS  1989    76.4    1850. Eighteen OECD Countries 1842.
# i 899 more rows
```

The Figure and some Tables

Now we write some code to draw the plot we want. The results are shown in [Figure 1](#).

```
df_plot |>
  ggplot(aes(
    x = avg_spend,
    y = life_exp,
    group = country,
```

```

    color = us_flag
  )) +
  geom_line(linewidth = 0.9) +
  scale_color_manual(values = my_colors) +
  scale_x_continuous(labels = scales::label_dollar()) +
  labs(
    color = NULL,
    title = "Health Spending and Life Expectancy, 1970-2023",
    x = "Heath Spending (Per capita, constant prices, constant PPPs, five year rolling average)",
    y = "Life Expectancy",
    caption = "Data: OECD. Graph: @kjhealy"
  ) +
  theme_bw() +
  guides(color = guide_legend(nrow = 1)) +
  theme(legend.position = "top", legend.text.position = "top")

```

Health Spending and Life Expectancy, 1970–2023

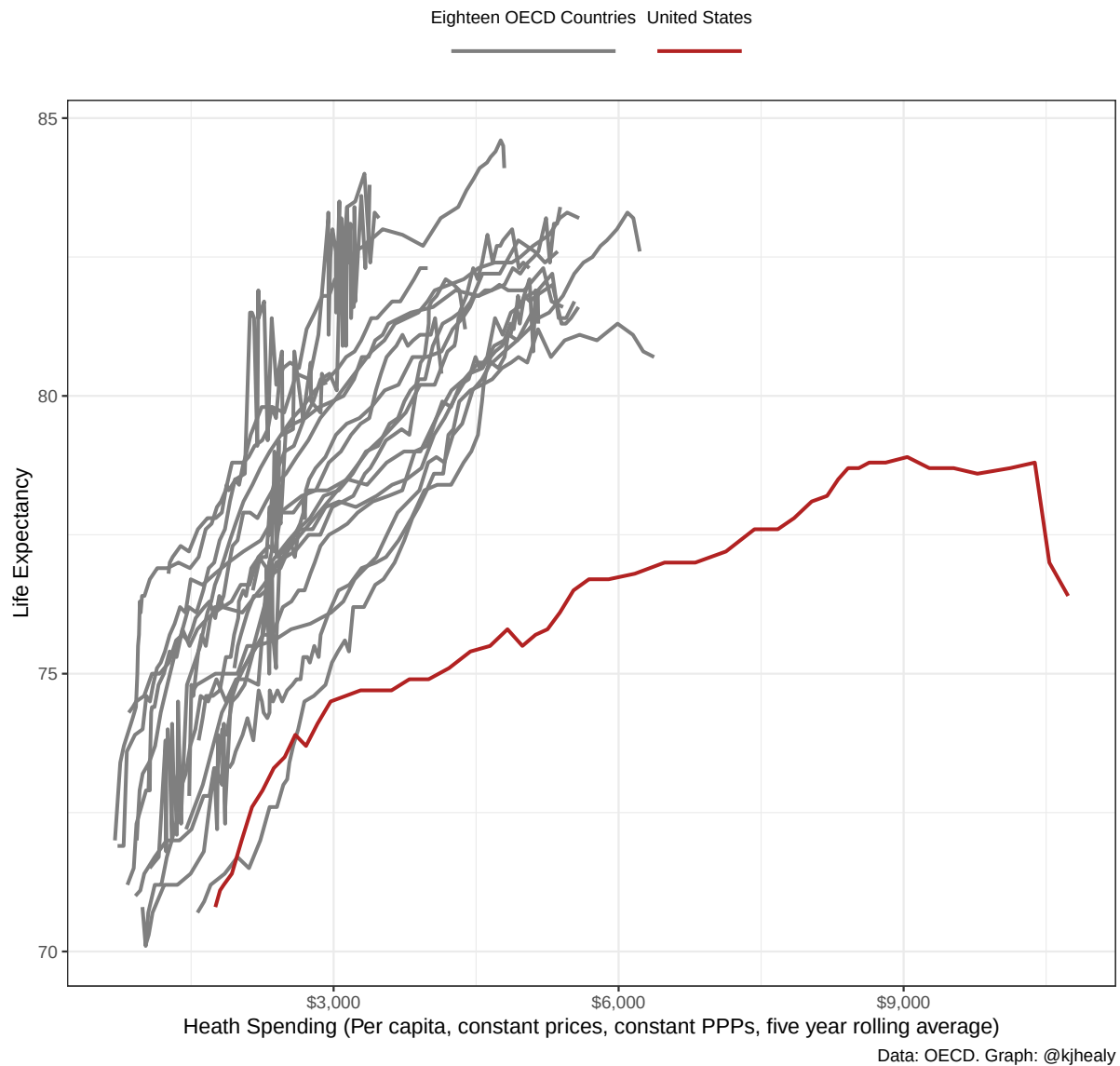


Figure 1: The figure we were trying to draw

Let's also make summary table or two while we are here. First, a table of the average life expectancy at birth for every country. This is shown in Table 1.

```
df_plot |>
  summarize(`Mean` = round(mean(life_exp), 1)) |>
  rename(Country = country) |>
  tt()
```

Table 1: Average Life Expectancy at Birth, in years, 1970-2023

Country	Mean
Australia	79.3
Austria	77.0
Belgium	77.0
Canada	79.2
Denmark	77.0
Finland	76.9
France	79.7
Germany	76.9
Greece	79.5
Ireland	77.5
Italy	80.7
Japan	79.7
Netherlands	78.3
New Zealand	77.0
Norway	78.5
Spain	79.0
Sweden	79.0
United Kingdom	77.5
United States	75.9

And second, Table 2 summarizes spending on health each year across countries.

```
df_plot |>
  group_by(year) |>
  summarize(across(
    health_ppp,
    list(
      Min = \(x) min(x),
      Mean = \(x) mean(x),
      Median = \(x) median(x),
      Max = \(x) max(x)
```

Table 2: Range of Spending across countries in Constant PPP per capita, selected years 1970-2023, rounded to the nearest dollar.

Year	Min	Mean	Median	Max
1970	466	962	906	1,663
1975	764	1,557	1,461	2,145
1980	936	1,749	1,774	2,666
1985	976	1,909	1,881	3,455
1990	1,121	2,275	2,413	4,470
1995	1,484	2,567	2,373	5,255
2000	1,904	3,081	2,796	6,068
2005	2,687	3,763	3,508	7,682
2010	2,964	4,282	4,234	8,489
2015	2,123	4,595	4,669	9,355
2020	2,348	5,102	5,171	11,081
2023	3,249	4,699	5,078	5,392

```

),
.names = "{fn}"
)) |>
mutate(across(
  starts_with("M"),
  \(x) scales::label_currency(accuracy = 1, prefix = "")(x)
)) |>
filter(year %in% c(seq(1970, 2023, 5), 2023)) |>
rename(Year = year) |>
tt()

```

References

Kenworthy, Lane. 2014. *Social Democratic America*. Oxford University Press.